ALGEBRA 1 · LINEAR INEQUALITIES

Inequality Number Line

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Inequality Number Line** on Mr. K's Classroom: Algebra 1 → Linear Inequalities → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Is 2 a solution of $x > 2$? Is it a solution of $x \geq 2$?

   No for $x > 2$; yes for $x \geq 2$.

2. $4 > 1$ is true. If you multiply both sides by -2, is $-8 > -2$ true?

   No. -8 is less than -2, so the statement becomes false.

2 Explore use the tool and record what you see

Graph and test

Set each inequality, then find one test point that makes it true.

Inequality	Open or closed circle?	Shade left or right?	A test point that works
$x > 3$	**open**	**right**	**any number above 3, like 5**
$x \leq -1$	**closed**	**left**	**−1 or less, like −4**
$x \geq 0$	**closed**	**right**	**0 or more, like 2**
$x < -2$	**open**	**left**	**below −2, like −5**

Why does the symbol flip?

Use the second tab. Start with a true statement and do the same thing to both sides.

Start (true)	Do to both sides	Result	Still true?
$4 > 1$	add 3	**$7 > 4$**	**yes**
$4 > 1$	multiply by 2	**$8 > 2$**	**yes**
$4 > 1$	multiply by -2	**$-8 > -2$**	**no**
$-3 < 2$	divide by -1	**$3 < -2$**	**no**
$4 > 1$	multiply by -2, then flip	**$-8 < -2$**	**yes**

3 Write the rule fill in each blank from what you observed

1. Use an open circle for **<** and **>**. Use a closed circle for **≤** and **≥**.

2. Adding or subtracting the same number on both sides **does not change** the inequality symbol.

3. Multiplying or dividing both sides by a positive number **does not change** the symbol.

4. Multiplying or dividing both sides by a negative number **flips (reverses)** the symbol, because it reverses the order of numbers on the number line.

4 Check no tool: show your work

1. Solve $-3x > 12$.

   $x < -4$

2. Solve $\frac{x}{-2} \leq 5$.

   $x \geq -10$

3. Solve $2x - 5 \geq 7$ and describe the graph.

   $x \geq 6$; closed circle at 6, shaded right

4. Solve $4 - x < 1$.

   $x > 3$

5 Reflect

Explain to a friend why $-2x < 6$ becomes $x > -3$, not $x < -3$.

Dividing by −2 reverses the order of the numbers, so the symbol must flip to keep the statement true. Test x = 0: −2(0) < 6 is true and 0 > −3 is true.